Phase 4 — The EIA Release Lab, Explained for Everyone

What happens to the price of oil at the exact moment the U.S. government publishes how much oil is sitting in the country's storage tanks — and can we call it in advance?

This guide assumes you know nothing about oil, trading, or statistics. Every special word is explained the first time it appears. Worked example: the 24 June 2026 EIA crude release.

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0. The one-sentence version

> Every week the U.S. government announces how much oil is in storage. Phase 4 is a little "laboratory" that, **before** the announcement, writes down a prediction of what the number will be and how the oil price should react — and then, **after** the announcement, checks whether it was right and *explains why*.

Think of it like a weather forecast that is graded against the actual weather, except the "weather" is the oil market's reaction to a government report.

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1. The five words you need (and nothing more)

Before anything else, here are the only pieces of jargon in this whole document. Each is plain once you see it:

- **Crude oil** — unrefined oil, straight out of the ground, before it is turned into petrol/ gasoline, diesel, jet fuel, etc.
- **Inventory / stocks** — the amount of crude oil currently sitting in storage tanks across the United States. Measured in **MMbbl = millions of barrels** (one barrel ~ 159 litres).
- **Draw** — stocks went **down** this week (oil was taken out of storage). Usually good for the price, because it hints demand is strong or supply is tight. A falling-supply story is called **bullish** (expecting prices to rise).
- **Build** — stocks went **up** this week (oil piled up in storage). Usually bad for the price — that's called **bearish** (expecting prices to fall).
- **WTI** — "West Texas Intermediate", the headline price of U.S. crude oil. When the news says "oil rose 2% today", this is usually the number they mean. (Its global cousin is **Brent**.)

That's it. Everything below is built from these five ideas.

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2. The single most important idea in the whole project

If you remember **one** thing, remember this:

> **The market does not react to the inventory number. It reacts to the surprise — how far the number lands from what everyone was already expecting.**

Why? Because traders are not surprised by things they already saw coming. If everyone *expects* oil stocks to fall by 4 million barrels, and they fall by exactly 4 million, the price barely moves — that outcome was already "baked into" today's price. The price only jumps when reality differs from the expectation.

A concrete, slightly counter-intuitive example:

- Stocks **fall** by 5 million barrels (a *draw*, normally good news for the price)…
- …but everyone expected them to fall by **10** million.
- Net effect: this is actually **bad** news — far less oil left storage than hoped. The price can *fall* on a draw.

So we define, very precisely:

> **Surprise = Actual number − Expected number.**

The Release Lab's entire job is to (1) build a sensible *expected* number, (2) measure the *surprise* once the real number lands, and (3) judge how much that surprise *should* move the price given the mood of the market that day.

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3. What actually gets published, and when

- **What:** every week the **EIA** (Energy Information Administration — the U.S. government's energy-statistics agency) publishes how much crude oil is in U.S. storage tanks, in a report called the *Weekly Petroleum Status Report*.
- **When: Wednesday at 10:30 a.m. New York time.** To the second. Traders are watching the clock.
- **Catch:** the weekly figure is an *estimate* (the EIA surveys most of the industry and fills in the rest), so treat it as a fast-but-slightly-noisy reading, not gospel.

This particular experiment is built around the release on **2026-06-24**, which reports on the week ending **2026-06-19**.

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4. The Lab in two halves: the Prediction and the Result

The page has a deliberate before/after structure. The left side ("**Expected**") is frozen *before* the number comes out. The right side ("**Real**") is filled in *after*, when you press the **Run** button. Then the Lab grades itself.

4a. The PREDICTION — written before the number lands

Everything here uses **only information available before the release** — it is never allowed to peek at the answer it is trying to predict. (In data science this is called being **leak-free**: no information "leaks" backwards from the future into the prediction.)

The expected number. Real Wall-Street forecasts are paywalled, so the Lab builds its own stand-in for "what everyone expects". It learns from roughly a dozen clues that are all known *the day before* the release — the *normal* change for this calendar week (the seasonal pattern), the last couple of weeks' changes (weeks tend to cluster), the recent supply/demand balance, and the lagged flow drivers: how hard refineries were running, production, imports vs exports, and last week's draws at **Cushing** and in petrol/diesel stocks (product draws often lead crude). These are blended with a **ridge regression** — a standard statistical recipe that weighs many clues at once while deliberately staying cautious so it doesn't over-fit the few hundred weeks of history. Crucially it is fit **walk-forward**: to predict any given week it may only use weeks *before* it. On the long backtest this beats the naive "just use the seasonal average" guess (see the Track Record below).

> For **24 June 2026 EIA crude release**, the model expected a **5.4 million-barrel draw** (-5.4 MMbbl).

The lean. This is the Lab's gentle *structural* bias — its background guess for which way the market is tilted, separate from the single weekly number. It can be **Bullish** (leaning toward higher prices), **Bearish** (lower), or **Neutral** (no strong tilt).

> The lean here was **Bullish** (confidence: low). In plain terms: oil stocks are unusually low for the time of year, they have been falling for weeks, and **Cushing** — the single town in Oklahoma where WTI oil is physically delivered, and therefore the spot that matters most for the WTI price — has been draining toward its lows. All of that leans toward higher prices.

The expected surprise is zero — on purpose. Because the model *expects the number to land on its own forecast*, its expected surprise is, by definition, about **0**. The whole game is what happens when reality misses that forecast.

The "catalyst strength" — the honest caveat. A **catalyst** is something that actually *causes* a price move. The Lab measures how strong a catalyst the inventory number has *been* in markets like today's, using a statistic called **R2** ("R-squared"):

> **R2 answers: "out of everything that moved the oil price on release day, what fraction was explained by the inventory surprise?"** It runs from 0 (the surprise explained *nothing* — the price moved entirely on other news) to 1 (the surprise explained *everything*).

> For this release the catalyst R2 was about **0.03** — essentially **zero**. Translation: in conditions like today's, the weekly inventory number has historically been a **weak** driver of the oil price. Other forces — economic news, OPEC (the group of big oil- exporting countries that coordinate production), wars and geopolitics — usually dominate the day.

This is the Lab being honest *up front*: it tells you, before the number even lands, that the number probably won't be the main event.

The impact curve and the scenarios. The Lab still draws the textbook relationship: a line saying "for each extra million barrels of surprise, WTI should move *this* much". The slope of that line is called **beta** (here about -0.083% per million barrels — a tiny number, consistent with the weak-catalyst finding). It then sketches three what-ifs:

If the surprise is…	…the model expects WTI to…
A big extra draw (much less oil than expected)	rise a little (+0.41%)
Exactly in line (no surprise)	barely move (-0.00%)
A big extra build (much more oil than expected)	fall a little (-0.41%)

Notice how *small* all three moves are — again, the weak-catalyst story.

4b. The RESULT — filled in after the number lands

When you press **Run**, the Lab fetches the real number and computes everything:

- **Actual:** stocks came in at a **6.1 million-barrel draw** (-6.1 MMbbl).
- **Real surprise:** -0.7 MMbbl. Recall surprise = actual – expected: a bigger draw than expected, which is a **bullish** (price-supportive) surprise.
- **How big is "big"?** We measure surprises in **sigma (s.d.)**, i.e. **standard deviations** — a statistician's ruler for "how unusual is this compared to a typical week?" Roughly: under 1s.d. is an ordinary week, 2s.d. is a genuinely large surprise. This one was **-0.1s.d.** — a small, ordinary-sized miss.

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5. Under the hood — the model, the prediction, and the logic

This is the engine room: exactly how the Lab turns a raw government statistic into a graded call, and the reasoning behind every step. Nothing here is a black box.

5a. The whole pipeline, in seven steps

> **Data -> Expected -> Surprise -> Regime -> Verdict -> Grade -> Track record.**

1. **Pull the data.** The weekly EIA report (crude stocks plus the flows behind them — production, imports, exports, refinery runs), recent prices, and the broad market.
2. **Predict the number (the model).** Estimate this week's draw/build *before* it is announced — the "Expected" figure (detailed in 5b).
3. **Measure the surprise.** When the real number lands: **Surprise = Actual - Expected.** Only this gap is genuinely *new* information.
4. **Read the regime.** Gauge how much the market is *listening* to inventories right now (calm vs. stormy market, the season, whether crude and fuels agree) — the "catalyst strength".
5. **Form the verdict.** Translate the surprise, scaled by the regime, into Bullish / Bearish / Neutral — alongside a slower **structural lean** from the multi-week backdrop (5d).
6. **Grade the outcome.** After the price moves, the scorecard checks the call and the attribution splits the move into inventory vs. macro vs. everything else.
7. **Repeat across history.** Run the identical logic on ~260 past releases (the track record), so nothing rests on a single lucky week.

5b. The model that predicts the number — ingredient by ingredient

The model's only job is to answer, *before* Wednesday's release: **how much will crude stocks change this week?** It learns from about a dozen clues, all known the day *before* the release — each chosen because there is a real-world reason it should help:

Clue the model uses	Why it helps predict this week's draw/build
Seasonal-normal change	demand and refinery schedules follow the calendar every year
Last 1-2 weeks' change	draws and builds arrive in runs (momentum)
Last week's supply/demand balance	physical flows are sticky from week to week
Refinery utilization & runs (lagged)	the harder refineries run, the more crude they pull from tanks
Crude production (lagged)	more barrels pumped tilts toward a build
Net imports (lagged)	imports add to storage; exports remove from it
Cushing change (lagged)	the WTI delivery hub often moves first
Gasoline & distillate draws (lagged)	strong fuel demand pulls crude through refineries
Stock level vs. seasonal norm (lagged)	unusually high or low stocks tend to revert
Week-of-year (sine/cosine)	smooth seasonality the weekly average alone misses

These are blended with a **ridge regression** — a standard, transparent way to weigh many clues at once. Every weight is learned only from the past (5c).

5c. Two ideas that keep the prediction honest

- **Ridge (don't over-trust any one clue).** With a dozen clues but only a few hundred weeks of history, a naive fit would start "memorising" random noise — looking brilliant on the past and then failing on the future (**overfitting**). Ridge deliberately *shrinks* the weights so no single clue dominates, trading a little fit on old data for reliability on new data.
- **Walk-forward (no peeking).** To predict any given week, the model may use **only the weeks before it**. So every "Expected" number is a genuine out-of-sample forecast, never hindsight. (This is the **leak-free** rule from earlier, applied across the whole history.)

5d. From a number to a call — the chain of reasoning

> Expected -> Surprise -> (scaled by) Regime -> Verdict, with a separate structural Lean.

- **Expected** is the consensus the price is *already positioned for* — so a print in line with it should barely move anything.
- **Surprise** (actual - expected) is the only part that is news. We size it in **sigma** (how unusual the miss is) rather than raw barrels, so "big" means the same thing in every season.
- **Regime** decides how loudly that news lands: the same surprise is worth more in a calm, inventory-driven market than in a war-driven one. The Lab measures this as the catalyst **R2**.
- **Verdict** = the direction of the surprise (a bigger draw -> bullish), *scaled down* by both how ordinary the surprise was and how weak the catalyst is. A large surprise in a deaf market still yields a muted, low-confidence call — on purpose.
- **Structural lean** is separate and slower: where stocks sit vs. the seasonal norm, the recent draw/build momentum, and the Cushing trend. It is the background tilt, not the day's trade.
- **The honesty built in:** when the catalyst R2 is near zero, the Lab actively says *don't trust the number today* — and counts being right about that as a success, not a dodge.

5e. Does the prediction actually work?

Two different questions, answered honestly and *separately*:

- **Does it forecast the number well?** Yes. On the long walk-forward backtest its average miss is **4.0 MMbbl vs 4.5 MMbbl** for the naive "just use the seasonal average" guess (lower is better) — and its week-to-week forecasts track the real draws and builds far more closely (their correlation with reality **nearly doubles** versus the seasonal guess).
- **Does forecasting the number mean predicting the price?** *Not necessarily* — and the Lab is blunt about it. The price reacts only to the *surprise*, and only when inventories are the day's catalyst; often they are not. So a good number-forecast and a small price reaction can both be true at once. Bridging that gap is exactly what the catalyst R2 and the move attribution are for.

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6. The Scorecard — three traffic lights and a verdict

The Lab boils the whole experiment down to three yes/no questions, each shown as a traffic light (green = yes, red = no, amber = unclear):

Question (plain English)	Result	What it means
1. Did we forecast the number well?	GOOD — PASS (green)	We were off by 0.7 MMbbl (-0.1s.d.) — a close forecast.
2. Did the surprise break the way we leaned?	YES — PASS (green)	lean Bullish vs a bullish beat. Our bullish lean and a bullish surprise agreed.
3. Did the oil price actually follow the surprise?	NO — FAIL (red)	WTI -3.9% vs +0.1% implied. The price went the *other* way.

The third light is the interesting one. The surprise said WTI should drift **up** about +0.06%. Instead WTI **fell 3.92%** on the day. On the surface that looks like the model failed.

It is the opposite of a failure — and here is the whole point of Phase 4:

> Framework worked as designed: it flagged inventories as a WEAK same-day catalyst (R2~0.03), and sure enough WTI ignored the surprise and moved on other forces (macro / OPEC / geopolitics).

In plain words: the Lab *told us in advance* (light by light) that the inventory number was a weak catalyst here (R2 ~ 0). So when the price ignored a perfectly good draw and tumbled 3.92% on other news, that wasn't a broken forecast — it was **exactly the world the Lab had warned about.** Knowing *when not to trust the signal* is itself the valuable output.

This is the single most important takeaway of the whole phase: **a good framework doesn't just make a call; it tells you how much to trust the call — and is right about its own limits.**

The receipts — splitting the move into inventory vs macro vs everything else

A fair question: if the inventory surprise barely moved the price, what *did*? The Lab now splits the release-day move into measurable parts (a technique called **attribution**): the piece explained by the crude **surprise**, the piece explained by the broad **market** that day (the S&P 500 and the US dollar — a "risk-on/risk-off" reading), and the **residual** — everything left over (oil-specific news, OPEC, geopolitics, positioning).

Part of the -3.92% day's move	Size	Plain meaning
Inventory surprise	+0.05%	what the crude number itself was worth
Broad market (macro)	+0.04%	risk-on/off + the dollar
Other / residual	-3.84%	**the dominant driver** — oil-specific other forces

So of the day's -3.92% move, the inventory number was worth only +0.05% and the broad market +0.04% — the rest, **-3.84%**, was other forces. That is the same message as the R-squared, now in plain percentage terms you can read straight off. (Looking at the *whole* report rather than crude alone — adding petrol, diesel and Cushing — nudges the explained share from R2 0.014 to 0.022; still small, but it confirms the day really was driven from outside the report.)

7. The minute-by-minute view ("Intraday reaction")

The headline number above is the move over the whole release day. The Lab also zooms in to the **intraday** level — *intra-day* simply means "within the same day", here minute by minute right after the 10:30 a.m. release.

It compares two lines on a chart:

- the **predicted** path (where the surprise said the price should go — a nearly flat line, since the catalyst was weak), and
- the **actual** path (where WTI really went, sampled every 5 minutes from live market data).

> What we saw: in the first hour WTI sagged to about **-0.56%** (at the 30-minute mark) — far from the flat, near-zero line the surprise implied — and then *climbed back* over the second hour. The realized wiggles had almost nothing to do with the inventory number; they were driven by other flows. The best any minute-horizon's R2 reached was about **0.03** — again, essentially zero explanatory power.

The lesson repeats at high resolution: on this day, the oil price and the inventory surprise were moving to different drums.

8. "Is this just luck?" — the Track Record

One experiment proves nothing — maybe the Lab got this one release right (or wrong) by chance. So the Lab also **backtests** the whole framework: it replays history and asks the same questions of *every past release*. (**Backtest** = test a method on past data, pretending you didn't know the outcome, to see if it would have worked.)

Three honest questions, scored over the last 52 weeks and over all history:

Question	Last 52 weeks	All history	How to read it
Surprise -> price: when the surprise was sizeable, how often did WTI move the way it implied?	55%	62%	50% is a coin-flip; higher is a real edge.
Lean: how often did our pre-release tilt call the surprise's direction?	54%	50%	Again vs. a 50% coin-flip.
Forecast accuracy (MAE): average miss of our expected number vs. a naive "just use the seasonal average" guess (lower is better)	4.1 vs 4.5	4.0 vs 4.5	Our model beats the naive guess.

(**MAE** = "mean absolute error" — just the average size of the miss, ignoring whether it was too high or too low.)

The picture is honest, not hyped: a **small but real** edge over the long run (better than a coin flip, and the forecast genuinely beats the naive seasonal guess), and a frank acknowledgement that in recent, geopolitics-dominated months the edge from the number alone has thinned toward a coin flip — precisely why the Lab leans on the *regime* read rather than the raw number.

9. Putting it all together — the story of 24 June 2026 EIA crude release

1. **Before** the release, the Lab predicted a **5.4 MMbbl draw** and leaned **Bullish** — while flagging that the number itself was a **weak catalyst** today.
2. **The number landed:** a **6.1 MMbbl draw** — a slightly *bigger* draw than expected, a **bullish** surprise of -0.7 MMbbl (-0.1s.d.).
3. **Two of three lights went green:** the forecast was close, and the surprise broke the bullish way we leaned.
4. **The third light went red:** WTI *fell* 3.92% instead of nudging up — because, exactly as warned, the day belonged to other forces (the 2026 Strait-of-Hormuz geopolitics, macro, positioning), not to the inventory number.
5. **Net verdict:** the framework worked *as designed* — it made a call **and** correctly told us how little to trust it that day.

> **The whole point of the Release Lab in one line:** it turns a confusing government data release into a clear, graded, before-and-after story — and, crucially, it knows the difference between "we were wrong" and "we correctly said this wouldn't matter today."

Appendix — honesty and limits (plain English)

- **No peeking.** Every number on the "Expected" side is built only from data available *before* the release; the historical backtest never lets a prediction see its own future. (This is the *leak-free* property.)
- **Our "expected" is a stand-in.** True analyst-consensus forecasts are paywalled, so we rebuild a transparent, reproducible proxy. It could be upgraded later to ingest a paid feed.
- **Why the price often ignores the number.** The headline statistic, R2, is usually small because a whole day of unrelated world news is mixed into each day's price move. The *directional* questions (did the price at least go the right way?) are the more trustworthy gauge, and they show a small, conditional edge.
- **The biggest caveat right now.** In mid-2026 the oil market is dominated by geopolitics, not by weekly inventories. The Lab *correctly* down-weights the number in this environment — which means the most important variable this month isn't in the inventory data at all. The Lab telling you *that* is the feature, not a bug.

Source for every figure above: the live Release-Lab snapshot the dashboard reads (`server/data/release_lab.json`), produced by `analytics/release_lab.py`. Generated 2026-06-25 02:56:47 EDT.